

1 Training Job

Sparse MoE Distributed
Training (3 DP × 4 Pipeline)

S0 S1 S2 S3



Healthy worker

Failed worker (rank r)

2 Failure Detection + Safe-Point Repair (R1)

Failure Event
event <r, t, c>

Repair Controller
(Safe-Point Guard)

- mark r RECOVERING
- block opt commit
- discard iteration

r : failed rank
t : failure step
(current)
c : checkpoint step

3 Guarded Policy (R2)

Policy Metric

$$\Phi'(t) = \frac{S(t) + |E_{\text{new}}| \cdot \Delta}{N_{\text{expert}} \cdot W}$$

Inputs

- $\Delta = t - c$
- peerAvail
- $S(t)$
- E_{new}

Thresholds

- $\Delta_{\min}$
- $\Delta_{\max}$
- $\Phi_{\max}$

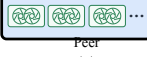
YES

guards pass?
HYBRID ?

NO

4a Hybrid Recovery (fast path)

Healthy peers
(step t)



Peer state

Replacement
Rank r'



Checkpoint Shard
(step c)



Checkpoint state

Two-Phase Recovery Protocol

Phase A:
weights first

resume training
under barrier

Phase B:
optimizer later

Reintegration (Guarded State Machine)

RECOVERING
(recovering r')

REPAIRED
(state synced)

BARRIER
(global sync)

HEALTHY
(fully active)

4b

Checkpoint Restart (fallback)

Full Checkpoint
(step c)

Reload
All Ranks

Replay
t - c

Resume
Training

5 Observability Logs (R3)



Auditable
recovery
decisions



policy inputs +
reason



TTTR / TTTFR
latency



state-machine
trace



quality metrics
(loss, ppl,
balance)

Control Flow

Path P (Peer)

Path C (Checkpoint)

Hybrid Path

Restart Path